# Evanns Morales-Cuadrado

U.S. Citizen

## Robotics PhD Candidate
## Fulbright Specialist
## Georgia Institute of Technology

**CONTACT**

**Email:** evannsmc@gmail.com
**Github:** https://github.com/evannsmc
**Research Website:** https://www.evannsmc.com/
**LinkedIn:** https://www.linkedin.com/in/evanns-morales/

---

**TECHNICAL SKILLS**

- **Programming Languages:** Python, C/C++, C#/.NET, Matlab/Simulink
- **Robotics & Control Frameworks:** ROS 2, NATS, PX4, ArduPilot, acados, CasADi
- **Machine Learning & Simulation:** JAX, PyTorch, MuJoCo, Reinforcement Learning
- **Development & DevOps Tools:** Docker, Continuous Integration, Git/GitHub
- **Hardware & Experimentation:** UAV assembly, repair, and deployment
- **Motion-Capture:** OptiTrack and Vicon
- **Languages:** Spanish (Native), English (Native), Portuguese (B1)

---

**EDUCATION**

**Georgia Institute of Technology** | 2022-2027
Ph.D. in Robotics
*Doctoral Minor in Optimization Methods*
- GPA: 4.0/4.0
- Advisor: Dr. Samuel Coogan
- Thesis Proposal Approved (ABD)
- Goizueta Fellow

**University of Texas at Arlington** | 2018-2022
Honors Bachelor of Science in Electrical Engineering
*Minor in Mathematics*
*Minor in Physics*
*Certificate in Unmanned Vehicle Systems*
- GPA: 3.9/4.0 Summa Cum laude
- Full National Merit Scholarship Awarded
- Electrical Engineering Honors Scholar

---

**WORK EXPERIENCE**

**Year-Round Graduate Research Intern** | AI for Autonomy — May 2026 - Present
Sandia National Laboratories — Albuquerque, NM
- Conducted applied research using reinforcement learning for autonomous systems
- Created novel robotics and automation infrastructure to advance national interests
- Developed C#/.NET networking frameworks and execution scripts for coordinating communication and task execution across multi-robot systems

**Teaching Assistant** | Vertically Integrated Project in Robotics — January 2024 - Present
Georgia Institute of Technology — Atlanta, GA
- Aided professor in his instruction of this research-based course
- Provided promising undergraduates with guidance performing applied research in the field of robotics
- Taught control theoretical concepts in an accessible manner for undergraduates

**Graduate Research Assistant** | FACTS Lab — August 2022 - Present
Georgia Institute of Technology — Atlanta, GA
- Advised by Dr. Samuel Coogan
- Implemented novel aggressive tracking control methods on quadrotor hardware
- Developed novel nonlinear control methods for safe autonomy
- Published novel research at top conferences and journals in the fields of robotics and control theory

**Research Assistant** | Autonomous Systems Lab — November 2021 - May 2022
University of Texas at Arlington — Arlington, TX

- Advised by Dr. Frank Lewis
- Received personal mentorship from a leading researcher in the field of controls and reinforcement learning (ranked $12^{th}$ in the USA and $23^{rd}$ in the world during my time in his lab)
- Aided graduate students on reinforcement learning research applied to quadrotor control

**Teaching Assistant** | Graduate-Level Intelligent Systems Course — January 2022 - May 2022
University of Texas at Arlington — Arlington, TX

- Aided professor in his instruction of the course
- Hosted review sessions for students in preparation for exams
- Gained teaching experience in a graduate-level course while still an undergraduate

**Research Assistant** | Dynamical Networks and Control Lab — October 2019 - May 2022
University of Texas at Arlington — Arlington, TX

- Advised by Dr. Yan Wan
- Gained experience with Robot Operating System (ROS), OpenCV, and path planning
- Original research in learning-based minimum time and energy path-planning for multi-vehicle systems

## SELECTED PUBLICATIONS

M. Cao, A. Harapanahalli, **E. Morales-Cuadrado**, and S. Coogan, "Trajectory Tracking for Systems with Unknown Time-Varying Disturbances." Submitted to Open Journal of Control Systems. *(under revision)*

**E. Morales-Cuadrado**, "Computationally Efficient and Safe Control for Aerial Robotic Systems under Threat and Disturbance." Ph.D. Thesis Proposal, Georgia Institute of Technology, 2026. *pdf available. slides available.*

**E. Morales-Cuadrado**, L. Chung, S. Kousik and S. Coogan, "`RTD-RAX`: Fast, Safe Trajectory Planning for Systems under Unknown Disturbances." Joint Submission to 2026 Modeling, Estimation and Control Conference (MECC) and ASME Letters in Dynamic Systems and Control (ALDSC). *(accepted to both) preprint available.*

**E. Morales-Cuadrado**, L. Baird, Y. Wardi and S. Coogan, "Lightweight Tracking Control for Computationally Constrained Aerial Systems with the Newton-Raphson Method." *IEEE Transactions on Control Systems Technology*, 2026. *preprint available.*

L. Baird, **E. Morales-Cuadrado**, and S. Coogan, "Runtime Assurance for Uncertain Systems from Interval Signal Temporal Logic." Submitted to IFAC Nonlinear Analysis: Hybrid Systems (NAHS), 2025. *(under revision) preprint available.*

**E. Morales-Cuadrado**, C. Llanes, Y. Wardi and S. Coogan, "Newton-Raphson Flow for Aggressive Quadrotor Tracking Control." *2024 American Control Conference (ACC)*

## RESEARCH INTERESTS

- Safe Autonomy
- Hardware Deployment
- Unmanned Ground and Aerial Vehicles
- Advanced Nonlinear Control
- Learning-Based Control
- Trajectory Generation and Planning

## OPEN-SOURCE CONTRIBUTIONS

- **PX4 Autopilot:** Contributed documentation improvements to the official PX4 Autopilot manual.
- **Vicon4PX4:** Developed and open-sourced a ROS 2 C++ Vicon–PX4 interface enabling external vision fusion for indoor quadrotor flight experiments; adopted and showcased by Georgia Tech's Indoor Flight Lab as part of its official codebase. Also created a parallel package for OptiTrack systems.

## SERVICE & LEADERSHIP

**Co-Founder and President of Puerto Rican Student Association** August 2024 - Present
Georgia Institute of Technology Atlanta, GA

- Addressed the need for an organization to help Puerto Rican students feel at home and stay connected to our culture
- Formed a three-member co-founder board and identified a faculty advisor
- Raised funds for the organization and recruited members from campus
- Hosted professional, social, and cultural events to meet the needs of the growing Puerto Rican student population at Georgia Tech

**Undergraduate Research Mentor** August 2021 - May 2022
University of Texas at Arlington Arlington, TX

- As the lead undergraduate student Dr. Frank Lewis' Autonomous Systems Lab, I mentored junior undergrads in controls research and guided them on a research project in ground vehicle control
- Guided students in Robot Operating System, Python, Linux, and Control Systems

**Teaching Assistant** | Graduate-Level Intelligent Systems Course January 2022 - May 2022
University of Texas at Arlington Arlington, TX

- Aided professor in his instruction of the course
- Hosted review sessions for students in preparation for exams
- Graded homeworks, exams, and projects
- Gained teaching experience in a graduate-level course while still an undergraduate

**Eta Kappa Nu Public Relations Chair & Circuits I Tutor** August 2020 - May 2022
University of Texas at Arlington Arlington, TX

- Executive board member of UT Arlington's IEEE-HKN chapter
- In charge of social media advertising for honor society's events and outreach initiatives for the electrical engineering student body
- Volunteered to provide free tutoring as a service to electrical engineering students in Circuits I

---

## HONORS AND AWARDS

**Fulbright Specialist** August 2026
Peer-review-selected to the U.S. Department of State's Fulbright Specialist Roster for a three-year term

**Top-3 Finalist at a Deep Learning Research Symposium** November 2023
Professor-sponsored award for novel research in deep learning and robotics

**Goizueta Fellowship at Georgia Tech** August 2022
Highly selective fellowship for graduate students of Latin-American descent

**Summa Cum Laude Honors at University of Texas at Arlington** May 2022
Exceptional academic distinction in top GPA tier for graduation honors

**Honors College Degree Requirements at University of Texas at Arlington** December 2021
Completed research projects in disparate fields throughout my undergraduate career with various professors and defended a senior capstone research project before the Honors College committee

**Electrical Engineering Honors Scholar at University of Texas at Arlington** December 2021
Inaugural member of the electrical engineering honors cohort

**IEEE-HKN International Electrical Engineering Honor Society Membership** August 2021
Selective undergraduate society recognizing outstanding academic achievement and community service

**Research Experience for Undergraduates Fellowship Recipient** August 2020
Fellowship to sponsor promising young undergraduate researchers

**Chance Vought Engineering and Science Endowment Scholarship** August 2020
Highly selective award given annually to three recipients across the entire College of Engineering

**Toray Composite Materials America Inc. Sons and Daughters Scholarship** May 2019
Academic scholarship awarded children of employees of Toray America

**James E. Casey Scholarship Recipient** January 2019
Academic scholarship awarded children of employees of UPS

**James R. Hoffa Memorial Scholarship Recipient** May 2018
Academic scholarship awarded children of Teamsters Union Members

**Grapevine, TX Rotary Club Scholarship Recipient** May 2018
Local rotary club scholarship for academic and civically-minded scholars

**National Hispanic Scholar** March 2018
Recognition of top national performers of Latin descent on the SAT

**National Merit Scholar** March 2018
Recognition of top national performers on the SAT